

Labs 2 and 3: Autoencoders, contrastive learning, and foundation models

The goal of this lab (and future labs) is to reflect the process of doing research. Therefore, the questions will be open-ended. There is no one right answer; instead, explain and justify the decisions you make as you do your analysis.

Important notes:

You can (and should) use a coding agent to assist you during your lab. Keep in mind that these agents are prone to shortcuts and hallucinations, so you must understand what is actually happening during your analysis. Throughout the lab, make notes on how you used a coding agent. Try to independently verify interpretations made by the agent by doing literature review. Another (sometimes, more robust) option is to start with validated findings from literature and explore them using the agent.

Final Submission

Submit a reproducible notebook and a corresponding PDF with all inline figures shown using [this](#) convertor. (Notebooks allow you to divide up code blocks with your own written findings/analysis and plot figures throughout. They are also easier for you to run, and for us to grade).

Within your notebook, answer all the following questions in order.

When relevant, include links and citations for both datasets and their associated papers, with a brief description of each dataset.

Part 1: Datasets

For Part 2: find one single-cell dataset and one spatial transcriptomics dataset associated with a published paper/preprint.

Make sure each dataset has a cell type annotation for each cell. This is because, in this lab, we will run this data through several different models and analyze the resulting embeddings by using the cell types as a point of reference.

For Part 3: find one single-cell dataset and one spatial transcriptomics dataset, which each contain data from multiple samples. For example, your datasets may have

- Several samples of the same tissue type (brain, tonsil, kidney, etc.) with the same annotated cell types across samples.
- Samples from multiple patients or other experimental conditions (healthy vs. diseased kidney, mouse liver exposed to antigen vs normal mouse liver, etc.)
- For spatial data: multiple adjacent 2-D tissue slices

In total, you should have 4 chosen datasets. For each, briefly describe:

- where you found the dataset
- the biological question studied in the associated paper
- the tissue/biological system studied
- the assay/technology used to collect the data
- the dataset dimensions
- any available annotations

Part 2: Models

Choose an established model for three different types of architectures discussed in class—(variational) autoencoders, contrastive learning, and foundation models—across your two data modalities (single-cell and spatial). That is, pick a model for each (modality, architecture) pair.

We provide some suggestions below but feel free to use a different model as long as you justify your choice; i.e. the model should be highly cited / well established or appear in a reputable journal.

Architecture	Single Cell	ST
VAE	scVI	STAGATE , DestVI
Contrastive Learning	Concerto , CLEAR , Contrastive-sc	GraphST
Foundation Model	scGPT , Geneformer , GenePT	Nicheformer , Novae

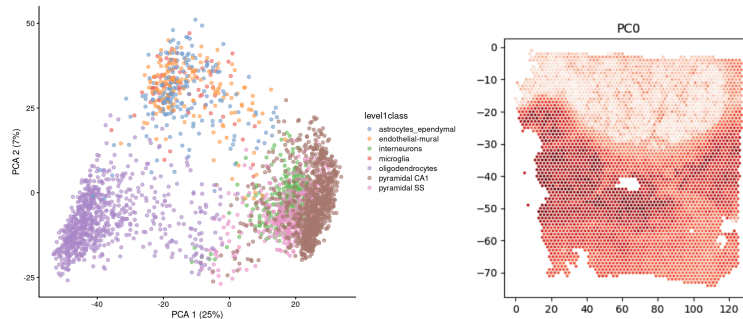
Part 2.1: Run models on individual samples

We will start by running each model M on an individual dataset D *without* replicates (i.e. the first two datasets you chose: the count matrix for single-cell data, and the count matrix + spatial coordinates for spatial data).

For a given dataset D and for each model M , do the following:

- Preprocess the dataset D according to the standard procedure detailed by the model M .
- Run the model M on the dataset D to learn embeddings E_M .
- Visualize the embeddings E_M in a manner that is most “useful” for understanding the data.
 - You may choose to visualize individual components of the embeddings E_M by visualizing the top PCs of E_M , or using another visualization method.
 - (Only use visualizations we have discussed in class (no t-SNE or UMAP).
 - For spatial data, you may wish to visualize individual embedding components or PCs using the 2-D physical coordinates.
 - In these visualizations, it is often helpful to label/color each individual cell by its cell type.

Below are two sample visualizations to give a sense of how to visualize the data. (Left: single cell, right: spatial)



Part 2.2: Comparing Embeddings across Models

We next want to compare the embeddings *across* different models M .

Pretend you are advising a biologist who just generated a bunch of really cool single cell/spatial data, and came to you asking which embedding model they should use.

- For all models applied to the dataset D , decide which model found the “best” embedding by using your visualizations.

You may define “best” however you want. Here is one suggestion:

- For each model M , do cells that are close together in its embedding E_M share the same cell type annotation?
 - You may quantify this by clustering the embeddings E_M and comparing the resulting cluster assignments with the provided cell type labels using ARI.
- For cell types that are known to be similar (e.g. two types of neurons), are they visually close in the embedding visualization?
 - For cell types that are known to be dissimilar (e.g. tumor versus normal cells), are they visually far apart in the embedding visualization?

Part 2.3: Training an MLP to predict cell type

We will next derive an approach to quantify how good embeddings are at predicting cell type.

For each model M trained on your dataset D :

- Train a standard MLP on the learned embeddings E_M to predict cell type. You can use any software, e.g. scikit-learn’s MLPClassifier, PyTorch, etc.
 - You should designate a subset of your cells as a “test set” and exclude this test set from training all of your MLPs.
- Compare the test accuracy of the MLP (i.e. how well the MLP does at predicting the cell type of the held-out test set) across all models M . Which embedding is best at predicting cell type, and why do you think this is the case?
 - Does this match your intuition from Part 2.2?

Part 3: Run models on all samples jointly

Next, run analyses on the data you chose with multiple samples $D_1, \dots, D_K$. Pick a model M which is designed to be trained on multiple samples (e.g. scVI).

Repeat parts 2.1, 2.2, and 2.3 by (1) training model M separately on each sample D_k and (2) *jointly* training model M on all samples $D_1, \dots, D_K$.

You should be able to

1. Visualize the embeddings for the model trained on each sample separately, and the embeddings of the jointly trained model.
 - a. Also try coloring the joint embeddings by sample of origin. Do the different samples appear to be integrated?
2. Compute the test accuracy of an MLP that uses the embeddings to predict cell type
 - a. It may be useful to hold out one sample D_k as a “test sample” for evaluating the different trained MLPs
3. Evaluate which model (individual, or jointly trained) is “better”

Ultimately, you should aim to answer: is it better to train model M individually on a single sample or jointly across all samples?

Extra credit: repeat this analysis with another model M' that can be trained on multiple samples, and compare the results for model M versus model M'

Part 4: UMAP and t-SNE

We next want to compare the embeddings learned by our models with two other dimensionality reduction methods discussed in class: UMAP and t-SNE. For each of your four datasets, repeat the analyses from Parts 2 and 3 using UMAP and t-SNE instead of the previously chosen models. You may first apply PCA to reduce the dimensionality of your input data.

You should be able to:

1. Visualize the resulting embeddings, coloring each cell by its cell type annotation.
2. Repeat the analyses from Parts 2.2 and 2.3 to compare these embeddings with those learned by your previous models.
3. For the multi-sample datasets, compare the embeddings obtained by applying UMAP/t-SNE to each sample individually versus all samples jointly.
 - a. For the multi-sample datasets, also try coloring the joint embeddings by sample of origin. Do the different samples appear to be integrated?
4. Compare the embeddings obtained from UMAP and t-SNE. Do they preserve similar cell type relationships? Are there differences in how the two methods separate or group cells?

Ultimately, you should aim to answer: How do UMAP and t-SNE compare with the learned embeddings from Parts 2 and 3? Does your answer change depending on whether you evaluate the embeddings visually or using cell type prediction accuracy?

Part 5: Final Reflection

Answer this how you see fit. Some possible directions include:

- How did results differ between the single-cell and ST datasets? Did incorporating spatial information change the embedding spaces in some interesting way?
- Were there any cool/weird findings? Anything interesting/thought provoking?